Relational database

Tables, keys, event and availability clocks

Related records at different granularities

Task contract

Root entity, horizon, origins and metric

Observed labels and masked forecast requests

Point-in-time boundary: source event time < query cutoff

Phase-censored inputs; mature observed-label history and eligible peer context

1. Staged relational search (S1)

Schema-derived programs

Depth 1-4; bounded repeated paths

Family combinations; base + CDF

Annotations and payload budgets

GraphReduce

Filter, annotate, reduce and join



Typed entity-grain feature frame

Pluggable learner

CatBoost: evaluated reference

TFMs via compatible adapters

Validate representation + learner

ROC-AUC (higher) / MAE (lower)

Validation drives promotion

10K screening → 50K confirmation → full-source rerank

2. Production-faithful finalist check (S2)

Bounded comparison with a smaller completed graph

Matched fitting lifecycle for classification and regression

3. Temporal training policy (S3)

Recent-only / pooled support / temporal blends

Walk-forward origins and matched validation seeds

4. Optional candidates evaluated with one shared temporal gate

Text · frame ensembles · target domain · zero-aware · object breadth

One incumbent; matched origins and seeds; independent applicability; uncertainty and safety checks

Freeze and fit

Graph plan + ordered schema + selected training policy + fitted learner

Replay on forecast requests with feature discovery disabled

Keyed predictions

Entity / cutoff order

Outer test targets are scored externally after predictions are saved.